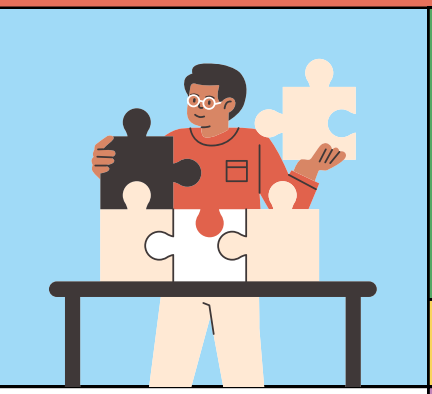


Master Builder's Brain Lab

Coding for joy, a science activity

Name: _____

Date: _____ Grade: _____

🚀 Let’s Start!

In this activity, you will learn about your brain and how it works. You will also learn about different kinds of joy. You will then use code in Canva to make a game.

UNDERSTAND THESE TERMS

- **Prompt Engineering:** Giving the AI the best "Magic Words" to get what you want.
- **Cheap Dopamine:** A fast, bored feeling from just watching (scrolling).
- **Earned Joy:** The proud feeling from finishing something hard (building).
- **Iterative Design:** The "Try-Again Loop" of testing and fixing your work.

Comprehension Tips

Read each question slowly. Think about what the words mean. Look back at the text if you need to. Ask a friend for help if you are stuck.

🧐 Audit Your Activities

Look at the activities below. Decide if they give you Quick Candy Joy or Hard-Work Happy. Write them in the right column. **THEN, write two of your own examples and state what they are.**

Activity	Is it **Hard-Work Happy (Earned Joy),** *or* **Quick Candy Joy (Cheap Dopamine)**
Watching TikTok	
Scrolling YouTube	
Building a LEGO set	
Finishing a Canva Game	

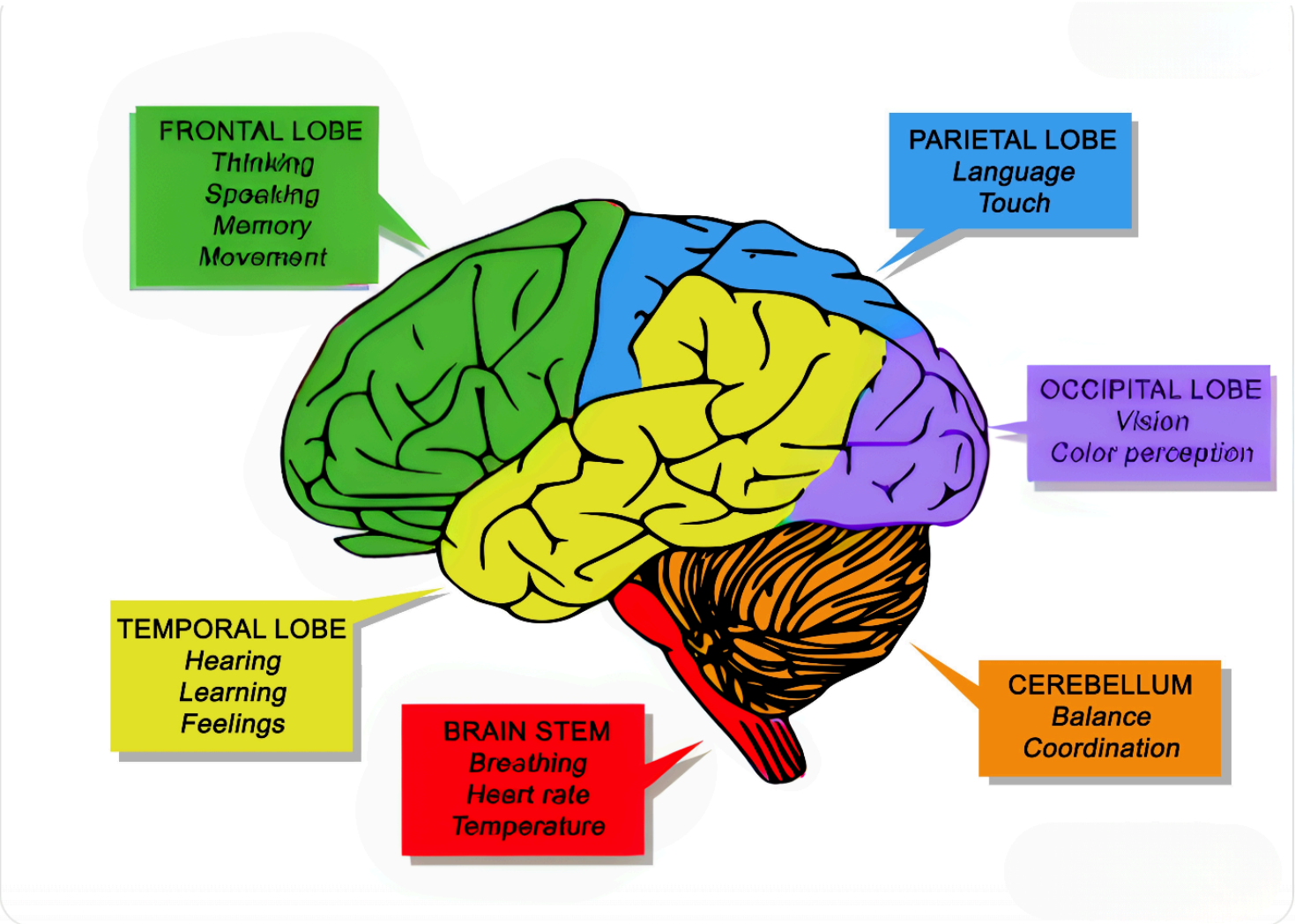
LEARN ABOUT the Parts of the Brain

Your Amazing Brain: A Guide to All the Parts

Your brain is like the control center of your body. It's made up of different parts that work together to help you think, move, feel, and stay alive. Let's explore each part and what it does!

Brain Parts Diagram

A side view of the brain showing all the major parts and their functions



Key:

- 🟢 **Frontal Lobe** (green) - Thinking, planning, personality
- 🔵 **Parietal Lobe** (blue) - Touch, spatial awareness
- 🟡 **Temporal Lobe** (yellow) - Hearing, memory, language

- 🟣 **Occipital Lobe** (purple) - Vision, visual processing
- 🟠 **Cerebellum** (orange) - Balance, coordination
- 🔴 **Brain Stem** (red) - Basic life functions

Word	Definition	What is YOUR Understanding?
Frontal Lobe (Green)	Thinking, planning, and your personality.	
Parietal Lobe (Blue)	Touch and knowing where things are.	
Temporal Lobe (Yellow)	Hearing, memory, and language.	
Occipital Lobe (Purple)	Seeing and understanding what you see.	
Cerebellum (Orange)	Balance and moving your body well.	
Brain Stem (Red)	Basic life functions, like breathing.	

🔑 Study Key Vocabulary

Let's learn some new words to help us understand how our brains work and how we can use them to build cool things.

Word	**Definition**	**What is YOUR Understanding?**
Prompt Engineering	Giving the AI the best "Magic Words" to get what you want.	
Cheap Dopamine	A fast, bored feeling from just watching (scrolling).	
Earned Joy	The proud feeling from finishing something hard (building).	
Iterative Design	The "Try-Again Loop" of testing and fixing your work.	

🤖 Code a Game Challenge

Try these game prompts in Canva AI. What cool things can you create?

- **The Ocean Quest:** A dolphin hero in a neon coral reef. IF the dolphin touches a plastic bottle, THEN the score goes down. IF it touches a pearl, THEN it levels up.

- **The Space Lab:** A robot in a gravity-free lab. IF the robot collects a battery, THEN it moves faster. IF it hits a magnet, THEN the screen shakes.
- **The Forest Mystery:** A squirrel in a pixel-art forest. IF the squirrel finds a golden acorn, THEN a secret door opens. IF it falls in the river, THEN it restarts at the beginning.

Which one did you choose?

_____

Why did you choose it?

_____

_____

_____

🤔 Think About What You Learned

Answer these questions to think about what you learned today.

What part of the brain helps you plan your game?

How does earned joy make you feel?

What is something hard that you want to build?